

Econometric Methods I

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Lecture from MGTECON 603(Lecturer: Guido Imbens)

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1 Probability, Statistics, ML, and Econometrics: Goals and Differences

1.1 Three Main Goals of Analyzing Data

- **Description:** Describing/Summarizing economic data
- **Prediction:** Predicting the future economic outcomes
- **Estimation of Causal Effects:** ‘Predicting’ the differences in outcomes under various policies or interventions

1.2 Examples of Descriptive Studies

1.2.1 Chetty, Hendren, Kline, and Saez (2016, QJE)

- Title: Where is the Land of Opportunity? The Geography of Intergenerational Mobility in the United States
- Author: Chetty, Hendren, Kline, and Saez (2016, QJE)
- Abstract: *First, we characterize the joint distribution of parent and child income at the national level. (...) Second, intergenerational mobility varies substantially across areas within the U.S.*
- Data: Income data from Tax Records
- Question: What is the conditional distn of a child’s income given their parent’s income across geography?
- Method: Regression of the *rank* of child’s income on the *rank* of parent’s income

1.2.2 Katz and Murphy (1992, QJE)

- Title: Changes in Relative Wages, 1963-1987: Supply and Demand Factors
- Author: Katz and Murphy (1992, QJE)
- Abstract: *Movements in the college wage premium over this period appear to be strongly related to fluctuations in the rate of growth of the supply of college graduates.*
- Data: Wages, education, and experience from the CPS
- Question: How did relative wages change over time for different education/experience groups?
- Method: Comparisons of the *quantiles* of earnings over time by education/experience groups

1.3 Examples of Prediction Studies

1.3.1 Campbell Harvey (1989, Financial Analysts Journal)

- Title: Forecasts of Economic Growth from the Bond and Stock Markets
- Author: Campbell Harvey (1989, Financial Analysts Journal)

- Data: Quarterly growth in real GDP, yield spread between long- and short-run annualized yields, S&P 500
- Question: What predicts growth rate? Does the yield spread do better than the stock market in predicting GDP growth? (a purely forecasting problem)
- Method: Linear Regression

1.3.2 Predicting Handwritten Digits(MNIST)

- Data: $Y_i \in \{0, 1, 2, \dots, 9\}$ is the digit label, $X_i \in \mathbb{R}^{784}$ is the pixel information(black/white) for 400 pixels
- Question: What is the most likely outcome for Y_i given $X_i = x$?
- Method: A bunch

We wouldn't need or want confidence intervals for this problem, since we are only interested in the most likely outcome and the accuracy of the prediction.

1.3.3 Netflix Challenge

- Data: Characteristics of the movies and the characteristics of the users(including past rental behavior)
- Goal: Recommend movie given the characteristics of the user and the movie

1.4 Examples of Causal Studies

1.4.1 Card and Krueger (1996, AER)

- Title: Minimum Wages and Employment: A Case Study of the Fast Food Industry in New Jersey and Pennsylvania
- Author: Card and Krueger (1996, AER)
- Abstract: *Comparisons of employment growth at stores in New Jersey and Pennsylvania (where the minimum wage was constant) provide simple estimates of the effect of the higher minimum wage.*
- Data: Employment data from Fast Food Restaurants in NJ and PA in early and late 1992
- Question: What is the *causal* effect of a minimum wage increase on employment in NJ?
- Method: Difference-in-differences regression

1.4.2 Imbens, Rubin, and Sacerdote (2001, AER)

- Title: Estimating the Effect of Unearned Income¹ on Labor Earnings, Savings, and Consumption: Evidence from a Survey of Lottery Players
- Author: Imbens, Rubin, and Sacerdote (2001, AER)

¹The unearned income here is the social security earnings from individuals who played and won the lottery, which is a guaranteed income for an almost lifelong period.

- Abstract: *This paper provides empirical evidence about the effect of unearned income on earnings, consumption, and savings.*²
- Data: Social security earnings from individuals who played and won the lottery.
- Question: What is the *causal* effect of lottery winnings on labor earnings?
- Method: Linear Regression

The lottery participants might have different characteristics from the non-lottery participants, which raises a question regarding **external validity**. To address this, the paper compares the individuals from the lottery data to CPS and looked at the income distribution. Here, the authors found that the distribution of earnings is similar to that of the CPS.

1.4.3 Aviv Nevo (2001, Econometrica)

- Title: Measuring Market Power in the Ready-to-Eat Cereal Industry
- Author: Aviv Nevo (2001, Econometrica)
- Abstract: *I estimate price-cost margins, but more importantly I am able empirically to separate these margins into three sources*
- Data: Quantities and prices of cereals for different brands, in different cities, over time.
- Question: Is this industry a classic example of a concentrated differentiated products industry in which price competition is approximately cooperative and rivalry is channeled into advertising and new product introduction?
- Method: *Structural Model* of Supply and Demand

1.5 Probability and Statistics

- Probability: Given a **fully specified**³ probability model, what are the probabilities of particular events / the probability of particular events conditional on other events?
- Statistics: Given the **realizations** of a random variable Y , drawn from a distribution F_θ , what can we say about the parameters θ ?
 - ◊ Y may be a vector of observations and θ may be a finite dimensional vector or an unknown function.
 - ◊ **Estimation:** What is a good estimate of θ given the realization of Y ?
 - ◊ **Testing:** Is $\theta = \theta_0$ consistent with the observation Y ?

²Guido: The original motivation of this paper was to estimate the impact of poverty on children's future income.

³Probability model is fully specified if it specifies the probability of every event in the sample space. (which implies having all the parameters and the functional form of the model known)

1.5.1 Estimation

Suppose we have a random sample $Y_1, \dots, Y_N$ drawn from a distribution $N(\theta, 1)$, and we want to estimate the parameter θ . What is a good estimate of θ ? The most natural estimator is the sample mean $\hat{\theta} = \frac{1}{N} \sum_{i=1}^N Y_i$, which is both the **maximum likelihood estimator** and the **least squares estimator**. In this case, $\hat{\theta}$ has the following properties:

- **Normality:** $\hat{\theta} \sim N(\theta, \frac{1}{N})$
- **Unbiasedness:** $\mathbb{E}(\hat{\theta}) = \theta$
- **Efficiency:** $\hat{\theta}$ is the most efficient estimator of θ among all unbiased estimators (we can't do any better than this)⁴

1.5.2 Testing

Why and when are we interested in testing?

- In the medical/physics settings, $\theta \neq \theta_0$ itself might be interesting.
- In the social science settings, we want to see the “magnitude” of the results, and whether we have enough evidence for a certain claim.
- In many of these cases, we don't want to switch to a different model arbitrarily.
- However, the general rule of $|t| > 2$ is completely arbitrary. In terms of decision-making, relying on tests is not a good idea.

1.6 Machine Learning

1.6.1 Linear Regression

Suppose we have a predictive linear model with large number of parameters K :

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_K X_{iK} + \varepsilon_i$$

where $X_{i1}, X_{i2}, \dots, X_{iK}$ are the features and $\beta_0, \beta_1, \dots, \beta_K$ are the parameters. Then, $\hat{Y}_{N+1}$ is not a very good estimator for either Y_{N+1} or $\mathbb{E}(Y_{N+1}|X_{N+1})$.

The typical solution is to **regularize** the estimator. For example,

$$\min_{\beta} = \sum_{i=1}^N (Y_i - \beta_0 - \beta_1 X_{i1} - \beta_2 X_{i2} - \dots - \beta_K X_{iK})^2 + \lambda \sum_{j=1}^K |\beta_j| \quad (1.6.1)$$

is the **lasso** estimator, which *shrinks* the coefficients towards zero. This leads to a better out-of-sample performance.

Going a little further, the LASSO regularization parameter λ is often chosen using cross-validation. For example, we can consider the **leave-one-out cross-validation (LOOCV)**:

Define

⁴NB: It touches the Cramér-Rao lower bound. This is a direct case of the Rao-Blackwell Theorem.q1s2

$$\beta^{(i)}(\lambda) = \arg \min_{\beta} \sum_{j \neq i} (Y_j - \beta_0 - \beta_1 X_{j1} - \beta_2 X_{j2} - \dots - \beta_K X_{jK})^2 + \lambda \sum_{k=1}^K |\beta_k| \quad (1.6.2)$$

based on least squares omitting the i -th observation. Then, the LOOCV error is given by:

$$Q(\lambda) = \frac{1}{N} \sum_{i=1}^N (Y_i - \hat{\beta}^{(i)}(\lambda)_0 - \hat{\beta}^{(i)}(\lambda)_1 X_{i1} - \hat{\beta}^{(i)}(\lambda)_2 X_{i2} - \dots - \hat{\beta}^{(i)}(\lambda)_K X_{iK})^2 \quad (1.6.3)$$

From this, we can choose the λ that minimizes the LOOCV error: $\hat{\lambda} = \arg \min_{\lambda} Q(\lambda)$.

1.6.2 Kernel Methods

Another thing that can be done is using nonparametric models ($\mathbb{E}[Y_i|X_i = x] = g(x)$). If X_i is low-dimensional, traditional researchers use **kernel** methods:

$$\hat{g}(x) = \sum_{i=1}^N K\left(\frac{X_i - x}{h}\right) Y_i / \sum_{i=1}^N K\left(\frac{X_i - x}{h}\right) \quad (1.6.4)$$

1.6.3 Neural Nets

However, Kernel methods do not work well if X_i is high-dimensional. In this case, **deep neural networks** are often used.

$$g(x) = \gamma_0 + \sum_{s=1}^S \gamma_s h\left(\sum_{j=1}^J \beta_{sj} x_j\right) \quad (1.6.5)$$

Equation 1.6.5 is an example of a **two-layer neural network**, where the *activation function* $h(\cdot)$ is fixed and we estimate $S \times J$ parameters β_{sj} and $S + 1$ parameters γ_s . The **ReLU** function is a popular activation function ($h(z) = \max(0, z)$) to introduce non-linearity.

1.7 What is Econometrics?

Econometrics is not solely about predictions, but also (primarily) about causal effects:

- What is the causal effect of raising the minimum wage on employment?
- What is the causal effect of getting more education on earnings?

These questions require dealing with **endogeneity, exogeneity, and identification**. For example, people who *choose* to get more education may do so precisely because they are more likely to be high-ability individuals, and their expected returns to education are higher.

1.7.1 Ed Leamer(1983, AER)

“The applied econometrician is like a farmer who notices that the yield is somewhat higher under trees where birds roost, and he uses this as evidence that bird droppings increase yields. However, when he presents this finding at the annual meeting of the American Ecological Association, another farmer in the audience objects that he used the same data but came up with the conclusion that moderate amounts of shade increase yields. A bright chap in the back of the room then observes that these two hypotheses are indistinguishable, given the available data. He mentions the phrase “identification problem,” which, though no one knows quite what he means, is said with such authority that it is totally convincing.”

Bottom Line: What makes econometrics different from other data analysis is that we are interested in identification. There are settings where we have a clear definition of “identification”, a lot of cases we don’t.

1.7.2 Classical Econometric Example: Supply and Demand

In the supply and demand model, we know that the prices are not set *randomly* but deliberately in response to demand. We cannot estimate a demand function by simply comparing sales when prices are high and when prices are low. Instead, we build a *model* for the behavior of buyers and sellers:

$$Q_d(P) = \alpha_0 + \alpha_1 P + \alpha_2 Y + \varepsilon_d \quad (1.7.1)$$

$$Q_s(P) = \beta_0 + \beta_1 P + \beta_2 W + \eta_s \quad (1.7.2)$$

We consider the following equation of the **equilibrium condition**:

$$Q_d(P) = Q_s(P)$$

2 Randomization and Fisher's Exact P-Values

2.1 Introduction to Causal Inference

There are three key notions underlying the general approach to causality:

1. **Potential Outcomes:** Each outcome corresponding to the various levels of a treatment or manipulation
2. **Multiple Units and Stability(SUTVA)**
3. **Assignment Mechanism:** Why some are exposed to the treatment and some are not(crucial for inferring causal effects and serves as an organizing principle)

2.2 The Fundamental Problem of Causal Inference

Suppose we have a binary outcome/manipulation/treatment/cause variable $W_i \in \{0, 1\}$ for all $i = 1, \dots, N$.

We associate each action with a potential outcome:

- $Y_i(0)$: The outcome for the i -th unit if they are *not* exposed to the treatment
- $Y_i(1)$: The outcome for the i -th unit if they are exposed to the treatment

We would want to compute the **unit-level causal effects**(i.e., the direct comparison of the two potential outcomes) for each unit. For example, $\tau_i := Y_i(1) - Y_i(0)$ or $\tau_i := Y_i(1)/Y_i(0)$.

However, we observe only one value of W_i , and therefore only one of the potential outcomes is ever observed:

$$Y^{obs} \equiv Y(W) = \begin{cases} Y_i(0) & \text{if } W_i = 0 \\ Y_i(1) & \text{if } W_i = 1 \end{cases} \quad (2.2.1)$$

$$Y^{mis} \equiv Y(1 - W) = \begin{cases} Y_i(1) & \text{if } W_i = 0 \\ Y_i(0) & \text{if } W_i = 1 \end{cases} \quad (2.2.2)$$

Such situation, the fact that it is impossible to observe all components of the causal effects(τ_i) is called the **fundamental problem of causal inference**.

2.3 Why potential outcomes are useful

Potential outcome notion is consistent with the way economists traditionally think about demand/supply or production functions:

$$q^d(p) \text{ and } q^s(p)$$

Some causal questions are more challenging, including estimating the causal effect of “immutable” characteristics on economic outcomes. One solution is to make manipulation precise. For example, Bertrand and Mullainathan (2004) do not manipulate the ethnicity itself, but they manipulate the “perceived” ethnicity

of the applicants. In such case and most causal questions, we must be exact about what the manipulation is. We would have to ask the following questions:

- Can we see causal effects here?
- Can we do experiments in this case?
- What's the treatment here?

2.4 Leveraging Multiple Units

Because we cannot learn much about causal effects from a single observed outcome, we typically rely on multiple units to learn about causal effects. This does not come without any assumptions. In some cases, one's potential outcomes may be *dependent* on the potential outcomes of others. For example, in agricultural fertilizer experiments, researchers have taken care to separate plots using “guard rows”, or unfertilized strips of land between fertilized areas. In other words, if we have **spillovers**, changing the treatment for one unit may affect the potential outcomes of other units; such spillover effects are the main focus in the peer effects / social interactions literature, but in most of our discussion, we will assume that the potential outcomes are independent of each other. i.e., We make the **Stable Unit Treatment Value Assumption (SUTVA)**:

$$Y_i(1), Y_i(0) \text{ are independent of } Y_j(1), Y_j(0) \text{ for all } i \neq j$$

This assumption is crucial for the identification of causal effects.

2.5 Assignment Mechanism

A key ingredient is how each individual came to receive the treatment level:

$$Pr(W|Y(1), Y(0), X) \tag{2.5.1}$$

We can partition the cases into the following three categories:

- **Random Assignment:** known, no dependence on $Y(1), Y(0)$
- **Unconfounded Assignment, Stratified Random Design:** unknown, no dependence on $Y(1), Y(0)$
- **Confounded Assignment:** unknown, possible dependence on $Y(1), Y(0) \rightarrow$ the most challenging case

In the case of random assignment, we can use the potential outcomes to identify the causal effects. In the case of non-random assignment, we need to make some assumptions to identify the causal effects.

2.6 Random Assignment

- Finite population with N units
- M units are selected at random to receive the treatment; the remaining $N - M$ units receive the control
- The assignment probability is given by:

$$Pr(\mathbf{W} = \mathbf{w}) = \binom{N}{M}^{-1} \quad (2.6.1)$$

- There are $\binom{N}{M}$ possible assignment vectors $\mathbf{w}$, **all equally likely**
- Special case of equal size treatment and control groups: $M = N/2$

cf) Bernoulli Trials

A slightly different setting from the one above is the **Bernoulli trials** with

$$Pr(W = w) = 2^{-N} \quad \forall w \in \{0, 1\}^N \quad (2.6.2)$$

This is a slightly *less attractive* setting than below – we might end up with highly imbalanced treatment and control groups, and in the worst case, we might not be able to identify the causal effects by ending up with a single treatment or control group.

2.7 Exact Tests

Moving away from the course material, let's first review the basics of hypothesis testing.

A test is a function of the data that maps the data to a decision to reject or not reject the null hypothesis. The term **exact test** refers to a hypothesis test whose **p-value** is exactly computable from the distribution of the test statistic under the null hypothesis. *This is different from the **asymptotic (large-sample) approximation** such as the normal, t , or chi-squared distribution.* We say that the p-value⁵ is **exact** in the sense that it is derived directly from the true null distribution. Formally,

$$p = \sum_{x \in \mathcal{X}} P_{H_0}(X = x) \quad (2.7.1)$$

where $\mathcal{X}$ is the set of all possible “extreme” cases of the data under the null hypothesis.

TBU: Can assuming the error term being exactly normally distributed and then running t-test on the regression coefficients be considered as an exact test?

2.8 Sharp Null Hypothesis

Given data from a completely randomized experiment, Fisher (1935) was interested in testing **sharp null hypotheses**—that is, null hypotheses under which all values of the potential outcomes for the units in

⁵Def: The sum of the probabilities of equi-likely or less likely events under the null hypothesis.

the experiment are either observed or can be specified. This is distinct from the null hypothesis that the average treatment effect is zero, which is not a sharp null hypothesis. The null of a zero average is a weaker hypothesis because the average effect of the treatment may be zero even if for some units the treatment has a positive effect, as long as for others the effect is negative.

For example, a **sharp null hypothesis** can be written as:

$$\begin{aligned} H_0 : Y_i(1) &= b_i \cdot Y_i(0) + c_i \quad \forall i \\ H_1 : Y_i(1) &\neq b_i \cdot Y_i(0) + c_i \quad \text{for some } i \end{aligned}$$

For now, let us focus on the hypothesis $H_0 : Y_i(1) = Y_i(0) \quad \forall i$. When is this sharp null with zero individual treatment effects more interesting than the null of a zero average treatment effect? If the change in outcome of any individual is important itself, then the sharp null is more interesting. However, if we want to know the overall direction of the treatment effect, then the null of a zero average treatment effect is more interesting.

2.9 Test Statistic

From now on, we will assume that the total number of units in our finite population is N and the total number of units that receive the treatment is fixed at M :

$$M = \sum_{i=1}^N W_i \tag{2.9.1}$$

In this **fixed M design**, we have

$$Pr_{\text{fixed}}(\mathbf{W} = \mathbf{w}) = Pr_{\text{fixed}}(\mathbf{W} = \mathbf{w} | M) Pr_{\text{fixed}}(M) \tag{2.9.2}$$

$$= Pr_{\text{fixed}}(\mathbf{W} = \mathbf{w} | M) = \binom{N}{M}^{-1} \tag{2.9.3}$$

A **test statistic** is a known function $T = T(\mathbf{W}; Y^{obs}, X)$ of the assignment vector $\mathbf{W}$, observed outcomes Y^{obs} , and *pretreatment* covariates X .

The stochasticity of T is solely due to the stochastic nature of the assignment vector $\mathbf{W}$, leading to the **randomization distribution**⁶ of T .

Using this distribution, we can compare the observed test statistic T^{obs} against its distribution under the null hypothesis to compute the p-value.

Question: What are the two ingredients for Fisher's exact test approach?

- The choice of the null hypothesis
- The choice of the test statistic

⁶i.e., the histogram of all values for the statistic from all possible ways the experimental units could have been randomly assigned to the treatment and control groups.

2.9.1 The Rank Statistic

An important class of statistics involves transforming the outcomes to **rank** statistics before considering the differences by treatment status. This improves robustness (against skewness and outliers) relative to difference-in-means without having to make an arbitrary choice such as taking logs; however, it must be noted that it is not a good idea to use it with data that is zero-inflated.

In practice, we also subtract $(N+1)/2$ from each to obtain a **normalized rank** statistic that has average zero in the population:

$$R_i(Y_1^{obs}, Y_2^{obs}, \dots, Y_N^{obs}) = \sum_{j=1}^N \mathbf{1}\{Y_j^{obs} < Y_i^{obs}\} + \frac{1}{2} \left(1 + \sum_{j=1}^N \mathbf{1}\{Y_j^{obs} = Y_i^{obs}\} \right) - \frac{N+1}{2} \quad (2.9.4)$$

Given the ranks R_i , an attractive test statistic is the difference in average ranks for the treated and control units:

$$T = \frac{\sum_i W_i R_i}{\sum_i W_i} - \frac{\sum_i (1 - W_i) R_i}{\sum_i (1 - W_i)} \quad (2.9.5)$$

This statistic is invariant to monotone transformations of the outcomes and is zero under the sharp null.

2.10 Exact Test Procedure

1. Specify the null hypothesis H_0

$$H_0 : Y_i(1) = Y_i(0) \quad \forall i \quad (2.10.1)$$

2. Choose the test statistic T

- **Difference-in-means:** $T = \frac{1}{N} \sum_{i=1}^N (W_i Y_i - (1 - W_i) Y_i)$
- **Difference-in-medians:** $T = \text{median}(W_i Y_i) - \text{median}((1 - W_i) Y_i)$
: robust against outliers, but not against zero-inflated data
- **Difference in average normalized ranks:** $T = \frac{1}{N} \sum_{i=1}^N \frac{\text{rank}(W_i Y_i) - \text{rank}((1 - W_i) Y_i)}{N}$
: robust against outliers, but not against zero-inflated data (though better than the difference-in-medians)

3. Compute the observed test statistic T^{obs}

4. Compute T for all other possible assignment vectors $\mathbf{w}$ (depends on the **randomization scheme**)

- If the population size N is large, it is computationally difficult to compute all possible assignment vectors $\mathbf{w}$
- Instead, we can consider a large enough number of permutations of the treatment assignment to approximate the randomization distribution. (Details in section 2.10.1)

5. Compute the p-value as the sum of the probabilities of all values of T that are as or more extreme than the observed test statistic T^{obs}

2.10.1 Computation of P-values with large N

The p-value calculations so far have been exact. With large N , it may be infeasible to calculate the test statistic for every value of the assignment vector. In this case we rely on *numerical approximations* to compute the p-value. Formally, we randomly draw N -dimensional vector with $N - M$ zeros and M ones from the set of assignment vectors K times. We then approximate the p-value for our test statistic by the fraction of these K statistics that are at least as extreme in absolute values as $|T^{obs}|$.

Algorithm 2.1 Exact Test Procedure (with numerical approximation)

```

Given  $N, M, Y^{obs}, T(\cdot)$ 
Initialize  $T^{sim}$ 
 $T^{obs} \leftarrow T(W, Y^{obs})$ 
 $K \leftarrow 100000$ 
for  $k \in 1, \dots, K$  do
     $\mathbf{w}_k \leftarrow$  random permutation of  $\{0, 1\}^N$ 
     $T_k^{sim} \leftarrow T(\mathbf{w}_k, Y^{obs})$ 
     $T^{sim} \leftarrow T(\mathbf{w}, Y^{obs}, X)$ 
end for
Return  $p\text{-value} = \frac{1}{K} \sum_{k=1}^K \mathbf{1}\{|T_k^{sim}| \geq |T^{obs}|\}$ 

```

W. The thought that larger $|T|$ are more unlikely, is this an assumption or a fact?

Not an assumption; we implicitly think that the size of the statistic is likely to be around zero when the null is true.

• p-value: the sum of the probability of equi-likely or less likely events under the null hypothesis.

the set of ^{all} possible events = $\mathbb{R}$ (i.e., T can fall into any $t \in \mathbb{R}$).

p-value

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Not to be confused with the *P-factor*.

In null-hypothesis significance testing, the **p-value**^[note 1] is the probability of obtaining test results at least as extreme as the result actually observed, under the assumption that the null hypothesis is correct.^{[2][3]} A very small p-value means that such an extreme observed outcome would be very unlikely under the null hypothesis. Even though reporting p-values of statistical tests is common practice in academic publications of many quantitative fields, misinterpretation and misuse of p-values is widespread and has been a major topic in mathematics and metascience.^{[4][5]}

$$\text{But, } \Pr(T=t) = \Pr(T=t^{\text{obs}}) = \left(\frac{1}{6}\right)^{-1}$$

for all t 's that are found.

Then, p-value = 1.

We have to look at $|T|$ and the likelihood of $\Pr(|T| \geq t^{\text{obs}})$.
Bottom Line: Aren't we implicitly having an assumption here? $\rightarrow$ this is the basis of equi-likely v less likely

"Repeated Sampling Thought Experiment"
 $\rightarrow$ No distributional assumptions.

So it's not that we're making assumptions here.

2.11 Takeaways

- Testing based on random sample is different from testing for the population. There is no approximation or further assumptions in this case, and the source of randomness is not from the random sampling scheme. It is only the randomization itself that helps us understand the data.
- Randomization-based p-values underlie tests for treatment effects.
- In practice, using approximations to p-values based on normal approximations to the distribution of the t-statistic is often similar to the exact p-values to the exact p-values based on difference-in-means test.

$\therefore$ The t-statistic should be normally distributed in large samples by CLT.

- With very skewed distributions rank-based tests are much better, but one needs to be careful regarding the presence of masspoints.

3 Estimating Average Treatment Effects in Randomized Experiments

FISHER–NEYMAN EXCHANGE

(N) “So long as the *average* (italics in original) yields of any treatments are identical, the question as to whether these treatments affect *separate* yields on *single* plots seems to be uninteresting and academic ... ”

(F) “... It may be foolish, but that is what the z [FEP] test was designed for, and the only purpose for which it has been used. ...”

(N) “... I believe Professor Fisher himself described the problem of agricultural experimentation formerly not in the same manner as he does now. ...”

(F) “... Dr. Neyman thinks another test would be more important. I am not going to argue that point. It may be that the question which Dr. Neyman thinks should be answered is more important than the one I have proposed and attempted to answer. I suggest that before criticizing previous work it is always wise to give enough study to the subject to understand its purpose. Failing that it is surely quite unusual to claim to understand the purpose of previous work better than its author.”

Guido's with Neyman most of the way:)

3.1 Neyman's Repeated Sampling Approach

While Fisher was developing his p-value calculations for **sharp null hypotheses**, Jerzy Neyman was focusing on methods for estimating the **average treatment effects**. His approach was to consider an estimator and derive its distribution(bias and variance) under **repeated sampling** by drawing from the randomization distribution of the assignment vector $\mathbf{W}$ (i.e., $\mathbf{W}$ is still stochastic). The potential outcomes $\mathbf{Y}(1), \mathbf{Y}(0)$ are still fixed in this thought experiment as well.

3.2 Unbiased Estimation of the Average Treatment Effect

Neyman was interested in the **population average treatment effect** defined as:

$$\tau \equiv \frac{1}{N} \sum_{i=1}^N \underbrace{(Y_i(1) - Y_i(0))}_{\text{deterministic(only one is observed)}} \quad (3.2.1)$$

Suppose that we observed data from a completely randomized experiment in which M units were assigned to the treatment and $N - M$ units were assigned to the control. Given randomization, the intuitive estimator for the average treatment effect is the difference-in-means estimator:

$$\hat{\tau} = \frac{1}{M} \sum_{i=1}^N Y_i(1) - \frac{1}{N - M} \sum_{i=1}^N Y_i(0) \quad (3.2.2)$$

where M and $N - M$ are the number of units assigned to the treatment and the control, respectively. This estimator is unbiased for the population average treatment effect under randomization:

$$\mathbb{E}[\hat{\tau}] = \mathbb{E}\left[\frac{1}{M} \sum_{i=1}^N Y_i(1) - \frac{1}{N - M} \sum_{i=1}^N Y_i(0)\right] \quad (3.2.3)$$

$$= \frac{1}{N} \sum_{i=1}^N \mathbb{E}\left[\frac{W_i Y_i(1)}{M/N} - \frac{(1 - W_i) Y_i(0)}{(N - M)/N}\right] \quad (3.2.4)$$

$$= \frac{1}{N} \sum_{i=1}^N \frac{\mathbb{E}[W_i] Y_i(1)}{M/N} - \frac{(1 - \mathbb{E}[W_i]) Y_i(0)}{(N - M)/N} \quad (3.2.5)$$

$$= \frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0)) = \tau \quad (3.2.6)$$

3.3 Variance of the Difference-in-means Estimator

In this section, we will derive the variance of the difference-in-means estimator under randomization and also create confidence intervals for the population average treatment effect. *The confidence interval part requires an appeal to the CLT for large sample normality.*

To calculate the variance of the difference-in-means, we need the **second and cross moments** of the random variable W_i :

$$\mathbb{E}[W_i^2], \quad \mathbb{E}[W_i W_j] \quad (3.3.1)$$

where W_i is the treatment indicator for the i -th unit.

- Second Moment: $\mathbb{E}[W_i^2] = \mathbb{V}(W_i) + \mathbb{E}[W_i]^2 = \frac{M}{N} \left(1 - \frac{M}{N}\right) + \left(\frac{M}{N}\right)^2 = \frac{M}{N}$
- Cross Moment: $\mathbb{E}[W_i W_j] = \Pr(W_i = 1) \cdot \Pr(W_j = 1 | W_i = 1) = \frac{M}{N} \cdot \underbrace{\frac{M-1}{N-1}}_{\text{fixed M design}}$

3.3.1 True Variance of the Difference-in-means Estimator

$$\mathbb{V}_{\mathbf{W}}(\hat{\tau} | \mathbf{Y}(0), \mathbf{Y}(1)) = \mathbb{V}_{\mathbf{W}} \left[\frac{1}{M} \sum_{i=1}^N W_i Y_i(1) - \frac{1}{N-M} \sum_{i=1}^N (1 - W_i) Y_i(0) | \mathbf{Y}(0), \mathbf{Y}(1) \right] \quad (3.3.2)$$

$$\begin{aligned} &= \frac{1}{M^2} \mathbb{V}_{\mathbf{W}} \left[\sum_{i=1}^N W_i Y_i(1) | \mathbf{Y}(0), \mathbf{Y}(1) \right] \\ &\quad + \frac{1}{(N-M)^2} \mathbb{V}_{\mathbf{W}} \left[\sum_{i=1}^N (1 - W_i) Y_i(0) | \mathbf{Y}(0), \mathbf{Y}(1) \right] \\ &\quad - \frac{2}{M(N-M)} \mathbf{Cov}_{\mathbf{W}} \left[\sum_{i=1}^N W_i Y_i(1), \sum_{i=1}^N (1 - W_i) Y_i(0) | \mathbf{Y}(0), \mathbf{Y}(1) \right] \end{aligned} \quad (3.3.3)$$

$$= \frac{N-M}{N} S_1^2 + \frac{M}{N(N-M)} S_0^2 + \frac{1}{N} (S_0^2 + S_1^2 - S_{01}^2) \quad (3.3.4)$$

$$= \frac{S_0^2}{N-M} + \frac{S_1^2}{M} - \frac{S_{01}^2}{N} \quad (3.3.5)$$

where

- $S_0^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i(0) - \overline{Y(0)})^2$
- $S_1^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i(1) - \overline{Y(1)})^2$
- $S_{01}^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i(1) - Y_i(0) - \tau)^2$

are the population variances of the potential outcomes and the population variance of the unit-level treatment effects. The problem here is that S_{01}^2 does not have an estimatable counterpart because we cannot obtain the first difference of the potential outcomes for each unit. However, if we assume that the treatment effect is additive/constant (i.e., $Y_i(1) = Y_i(0) + \tau$), then $S_{01}^2 = 0$ and the true variance of the difference-in-means estimator is given by:

$$\mathbb{V}_{\mathbf{W}}(\hat{\tau} | \mathbf{Y}(0), \mathbf{Y}(1)) = \frac{S_0^2}{N-M} + \frac{S_1^2}{M} \quad (3.3.6)$$

3.3.2 Variance Estimator

Under the assumption of the additive treatment effect, we can use the following variance estimator:

$$\hat{\mathbb{V}}_{\mathbf{w}}(\hat{\tau} | \mathbf{Y}(\mathbf{0}), \mathbf{Y}(\mathbf{1})) = \frac{s_0^2}{N - M} + \frac{s_1^2}{M} \quad (3.3.7)$$

where s_0^2 and s_1^2 are the sample variances of the control and treatment groups, respectively.

This estimator for the variance is widely used, even when the assumption of an additive treatment effect is inappropriate. There are two reasons for this:

1. Confidence intervals generated using this estimator of the variance will be more *conservative* with actual coverage at least as large as, but not necessarily equal to, the nominal coverage.
2. In terms of the superpopulation variance, this estimator is unbiased.

3.4 Superpopulation Justification for the Neyman Variance Estimator

Suppose our sample is a **random sample** from a **superpopulation** of n units.

Then, the **superpopulation average treatment effect** is given by

$$\tau^P \equiv \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0)) \quad (3.4.1)$$

We also have sampling indicator $R_i = 0, 1$ for each unit i in the superpopulation, where $N = \sum_{i=1}^n R_i$ is the total number of units in our sample. Our **sample average treatment effect**⁷ is given by:

$$\tau^S = \frac{1}{N} \sum_{i=1}^n R_i \cdot (Y_i(1) - Y_i(0)) \quad (3.4.2)$$

Our difference-in-means estimator will be written as

$$\hat{\tau} = \frac{1}{M} \sum_{i=1}^N W_i R_i Y_i(1) - \frac{1}{N - M} \sum_{i=1}^N (1 - W_i) R_i Y_i \quad (3.4.3)$$

Also, let us define the following quantities in table ??.

What we had before was the variance relative to the sample mean τ^S . That is,

$$\mathbb{V}^S(\hat{\tau} | R_1, \dots, R_n) = \mathbb{E} \left[(\hat{\tau} - \tau^S)^2 | R_1, \dots, R_n \right] = \frac{S_0^{2,S}}{N_1} + \frac{S_1^{2,S}}{N_0} - \frac{S_{01}^{2,S}}{N} \quad (3.4.4)$$

First,

- $\mathbb{E}[R_i] = \frac{N}{n}$
- $\mathbb{E}[R_i^2] = \frac{N}{n}$

⁷Although formerly referred to as the **population average treatment effect**

	Control	Treatment
Superpopulation mean	$\bar{Y}(1)^P = \frac{1}{n} \sum_{i=1}^n Y_i(1)$	$\bar{Y}(0)^P = \frac{1}{n} \sum_{i=1}^n Y_i(0)$
Sample mean	$\bar{Y}(1)^S = \frac{1}{N} \sum_{i=1}^n R_i Y_i(1)$	$\bar{Y}(0)^S = \frac{1}{N} \sum_{i=1}^n R_i Y_i(0)$
Superpopulation variance	$S_1^{2,P} = \frac{1}{n-1} \sum_{i=1}^n (Y_i(1) - \bar{Y}(1)^P)^2$	$S_0^{2,P} = \frac{1}{n-1} \sum_{i=1}^n (Y_i(0) - \bar{Y}(0)^P)^2$
Sample variance	$S_1^{2,S} = \frac{1}{N-1} \sum_{i=1}^n R_i (Y_i(1) - \bar{Y}(1)^S)^2$	$S_0^{2,S} = \frac{1}{N-1} \sum_{i=1}^n R_i (Y_i(0) - \bar{Y}(0)^S)^2$
Superpopulation cross-variance	$S_{01}^{2,P} = \frac{1}{n-1} \sum_{i=1}^n \left(Y_i(1) - Y_i(0) - \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0)) \right)^2$	
Sample cross-variance	$S_{01}^{2,S} = \frac{1}{N-1} \sum_{i=1}^n R_i \left(Y_i(1) - Y_i(0) - \frac{1}{N} \sum_{i=1}^n R_i \cdot (Y_i(1) - Y_i(0)) \right)^2$	

Table 1: Superpopulation and Sample Quantities

If we take the unconditional expectation of the above equation, we get the following:⁸

$$\mathbb{E}[\mathbb{V}^S(\hat{\tau} \mid R_1, \dots, R_n)] = \mathbb{E}\left[\frac{S_0^{2,S}}{N_1} + \frac{S_1^{2,S}}{N_0} - \frac{S_{01}^{2,S}}{N}\right] \quad (3.4.5)$$

$$\begin{aligned} &= \mathbb{E}\left[\frac{\frac{1}{N-1} \sum_{i=1}^n R_i (Y_i(0) - \bar{Y}(0)^S)^2}{N_1}\right] + \mathbb{E}\left[\frac{\frac{1}{N-1} \sum_{i=1}^n R_i (Y_i(1) - \bar{Y}(1)^S)^2}{N_0}\right] \\ &\quad - \mathbb{E}\left[\frac{\frac{1}{N-1} \sum_{i=1}^n R_i (Y_i(1) - Y_i(0) - \tau^S)^2}{N}\right] \end{aligned} \quad (3.4.6)$$

$$= \frac{S_0^{2,P}}{N_1} + \frac{S_1^{2,P}}{N_0} - \frac{S_{01}^{2,P}}{N} \quad (3.4.7)$$

Now, let us consider the **superpopulation variance of the Difference-in-means estimator**:

⁸Note that we do still condition on the sample size and the number of units in the treatment and control groups. (N, N_1, N_0) .

$$\mathbb{V}^P(\hat{\tau}) = \mathbb{E}[(\hat{\tau} - \tau^S + \tau^S - \tau^P)^2] \quad (3.4.8)$$

$$= \mathbb{E}[(\hat{\tau} - \tau^S)^2] + 2\mathbb{E}[(\hat{\tau} - \tau^S)(\tau^S - \tau^P)] + \mathbb{E}[(\tau^S - \tau^P)^2] \quad (3.4.9)$$

$$= \mathbb{E}[\mathbb{V}^S(\hat{\tau})] + 2\mathbb{E}[(\tau^S - \tau^P)\mathbb{E}[\hat{\tau} - \tau^S | \tau^S]] + \mathbb{V}^P(\tau^S) \quad (3.4.10)$$

$$= \mathbb{E}[\mathbb{V}^S(\hat{\tau})] + 2\mathbb{E}[(\tau^S - \tau^P) \cdot 0] + \mathbb{V}^P(\tau^S) \quad (3.4.11)$$

$$= \mathbb{E}[\mathbb{V}^S(\hat{\tau})] + \mathbb{V}^P(\tau^S) \quad (3.4.12)$$

$$= \mathbb{E}\left[\frac{S_0^{2,S}}{N_1} + \frac{S_1^{2,S}}{N_0} - \frac{S_{01}^{2,S}}{N}\right] + \underbrace{\mathbb{E}\left[\frac{1}{N^2} \sum_{i=1}^N (Y_i(1) - Y_i(0) - \tau^P)^2\right]}_{\text{infinite superpopulation case}} \quad (3.4.13)$$

$$= \frac{S_0^{2,P}}{N_1} + \frac{S_1^{2,P}}{N_0} - \frac{S_{01}^{2,P}}{N} + \frac{1}{N^2} \mathbb{E}\left[\sum_{i=1}^N (Y_i(1) - Y_i(0) - \tau^P)^2\right] \quad (3.4.14)$$

$$= \frac{S_0^{2,P}}{N_1} + \frac{S_1^{2,P}}{N_0} - \frac{S_{01}^{2,P}}{N} + \underbrace{\frac{1}{N^2} \frac{(n-1)N}{(n-1)n} \mathbb{E}\left[\sum_{i=1}^n (Y_i(1) - Y_i(0) - \tau^P)^2\right]}_{\text{iid random sampling}} \quad (3.4.15)$$

$$= \frac{S_0^{2,P}}{N_1} + \frac{S_1^{2,P}}{N_0} - \frac{S_{01}^{2,P}}{N} + \frac{1}{N} \left(1 - \frac{1}{n}\right) \mathbb{E}\left[\frac{1}{n-1} \sum_{i=1}^n (Y_i(1) - Y_i(0) - \tau^P)^2\right] \quad (3.4.16)$$

$$= \frac{S_0^{2,P}}{N_1} + \frac{S_1^{2,P}}{N_0} - \frac{S_{01}^{2,P}}{N} + \frac{1}{N} \left(1 - \frac{1}{n}\right) S_{01}^{2,P} \quad (3.4.17)$$

$$= \frac{S_0^{2,P}}{N_1} + \frac{S_1^{2,P}}{N_0} - \frac{1}{n} S_{01}^{2,P} \quad (3.4.18)$$

$$\rightarrow \frac{S_0^{2,P}}{N_1} + \frac{S_1^{2,P}}{N_0} \quad \text{as } n \rightarrow \infty \quad (3.4.19)$$

Sarah: We're arguing that there is no heterogeneity of the treatment effects at the population level. Also, if we think about infinitely superpopulation, we do not need to use R_i sample indicators, but if we think about finite superpopulation, we need to use R_i sample indicators.

Thus, the Neyman superpopulation variance of the difference-in-means estimator does not have the third term (i.e., the treatment effect variance term), which validates dropping the third term in the Neyman variance estimator.

3.4.1 Superpopulation Unbiasedness of Difference-in-means Estimator

Since we know that the difference-in-means estimator is unbiased for the sample ATE, it suffices to show that the sample ATE is unbiased for the superpopulation ATE – applying the law of iterated expectations to the difference-in-means estimator gives us the desired result.

$$\mathbb{E}[\tau^S] = \mathbb{E}\left[\frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0))\right] \quad (3.4.20)$$

$$= \frac{1}{N} \sum_{i=1}^N \mathbb{E}\left[(Y_i(1) - Y_i(0))\right] = \tau^P \quad (3.4.21)$$

3.5 Confidence Intervals

3.5.1 Aside: Lyapounov Central Limit Theorem

The Lyapounov Central Limit Theorem (LCLT) is a generalization of the classical Central Limit Theorem that provides conditions under which the normalized sum of independent (but **not necessarily identically distributed**) random variables converges in distribution to a normal distribution. Specifically, as long as the Lyapounov condition on the moments of the random variables is met, the LCLT guarantees that the sum behaves approximately normal for large samples.

Theorem 3.5.1 (Lyapounov Central Limit Theorem). *Let $X_1, \dots, X_n$ be a sequence of independent random variables, each with finite first and second moments $\mu_i = \mathbb{E}[X_i]$ and $\sigma_i^2 = \mathbb{E}[(X_i - \mu_i)^2]$, respectively. Define the sum of the variances as $s_n^2 = \sum_{i=1}^n \sigma_i^2$.*

*If for some $\delta > 0$ (**Lyapunov condition**),*

$$\lim_{n \rightarrow \infty} \frac{1}{s_n^{2+\delta}} \sum_{i=1}^n \mathbb{E}\left[|X_i - \mu_i|^{2+\delta}\right] = 0 \quad (3.5.1)$$

then a central limit theorem holds:

$$\frac{1}{s_n} \sum_{i=1}^n (X_i - \mu_i) \xrightarrow{d} N(0, 1) \quad (3.5.2)$$

as $n \rightarrow \infty$.

The related approximation is:

$$\sum_{i=1}^n X_i \approx N\left(\sum_{i=1}^n \mu_i, s_n^2\right) \quad (3.5.3)$$

as $n \rightarrow \infty$.

3.5.2 Confidence Intervals for 50:50 Randomization

Suppose we have a completely randomized experiment with 50:50 randomization, and define $D_i = 2W_i - 1$ as the recentered and normalized treatment indicator. In other words,

$$\mathbb{E}[D_i] = 0 \quad \text{and} \quad D_i^2 = 1 \quad (3.5.4)$$

From now on, we ignore the random sampling from the superpopulation, but consider the experiment participants the population and N the population size (and thus, $\tau \equiv \tau^S$). Then, the difference-in-means estimator can be written as:

$$\hat{\tau} = \tau + \frac{1}{N} \sum_{i=1}^N D_i(Y_i(1) - Y_i(0)) \quad (3.5.5)$$

$$\sqrt{N}(\hat{\tau} - \tau) = \frac{1}{\sqrt{N}} \sum_{i=1}^N D_i(Y_i(1) + Y_i(0)) \quad (3.5.6)$$

Then, the whole term of $\sqrt{N}(\hat{\tau} - \tau)$ has mean zero and variance $\mathbb{V} = \frac{1}{N} \sum_{i=1}^N (Y_i(1) + Y_i(0))^2$. We can construct the confidence interval by applying Lyapunov CLT:

$$\frac{\sqrt{N}(\hat{\tau} - \tau)}{\sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i(1) + Y_i(0))^2}} \xrightarrow{d} N(0, 1) \quad (3.5.7)$$

3.6 Stochastic Estimands

We can further consider the following estimands:

- **Average Effect for the Treated:** $\tau_t = \mathbb{E}[Y_i(1)|W_i = 1] = \frac{1}{N_1} \sum_{i=1}^N W_i(Y_i(1) - Y_i(0))$
- **Average Effect for the Control:** $\tau_c = \mathbb{E}[Y_i(0)|W_i = 0] = \frac{1}{N_0} \sum_{i=1}^N (1 - W_i)(Y_i(1) - Y_i(0))$

It has to be noted that these estimands are *stochastic* because they are conditional on the treatment indicator W_i . However, the difference-in-means estimator is still unbiased for these estimands (i.e., $\mathbb{E}[\hat{\tau} - \tau_t] = 0$ and $\mathbb{E}[\hat{\tau} - \tau_c] = 0$).

3.6.1 Smaller Variance

$$\mathbb{V}(\hat{\tau} - \tau) = \mathbb{E}[(\hat{\tau} - \tau_t + \tau_t - \tau)^2] \quad (3.6.1)$$

$$= \mathbb{E}[\hat{\tau} - \tau_t]^2 + 2\mathbb{E}[(\hat{\tau} - \tau_t)(\tau_t - \tau)] + \mathbb{E}[(\tau_t - \tau)^2] \quad (3.6.2)$$

$$= \mathbb{E}[\hat{\tau} - \tau_t]^2 + \mathbb{E}[(\tau_t - \tau)^2] \quad (3.6.3)$$

and thus, we have

$$\mathbb{E}[(\hat{\tau} - \tau_t)^2] < \mathbb{E}[(\hat{\tau} - \tau)^2] \quad (3.6.4)$$

4 Least Squares Estimation in Randomized Experiments

In the Neyman world, we had the following settings and results:

- setting: completely randomized experiments
- Assumptions: $W_i \perp Y_i(0), Y_i(1)$
- Estimator: $\hat{\tau} = \bar{Y}_T - \bar{Y}_C$
- True Variance: $\mathbb{V}(\hat{\tau}) = \frac{1}{N_1} S_1^2 + \frac{1}{N_0} S_0^2 - \frac{1}{N} S_{01}^2$
- Estimated Variance: $\hat{\mathbb{V}}(\hat{\tau}) = \frac{1}{N_1} s_1^2 + \frac{1}{N_0} s_0^2$

In the OLS world, we have the following model, assumptions, and the estimator:

- model: $Y_i = \alpha + \beta W_i + \varepsilon_i$
- assumptions: $W_i \perp \varepsilon_i$ $\varepsilon_i \sim N(0, \sigma^2)$, implying $\mathbb{E}[Y_i|W_i] = \alpha + \beta W_i$
 - ◊ The mean zero assumption for the error term is not restrictive, as we can always center the outcome variable by subtracting the mean of the outcome variable using the constant term α .
 - ◊ The variance assumption is restrictive, as it implies that the error term is homoskedastic; however, we can relax this assumption by allowing heteroskedasticity.
- estimator: $\hat{\beta} = \frac{\sum_{i=1}^N (W_i - \bar{W})(Y_i - \bar{Y})}{\sum_{i=1}^N (W_i - \bar{W})^2}$

The assumption of $W_i \perp \varepsilon_i$ is more difficult to evaluate directly, as the interpretation of these residuals is rarely made explicit beyond a somewhat vague notion of capturing unobserved factors affecting the outcomes of interest.

Let α be the population average outcome under the control, $\alpha = \mu_0 = \mathbb{E}[Y_i(0)]$. Recall that τ is the super-population average treatment effect, and so we have $\tau = \mu_1 - \mu_0$. Now define the residual ε_i in terms of the population parameters, treatment indicator, and the potential outcomes as

$$\varepsilon_i = Y_i(0) - \alpha + W_i \cdot (Y_i(1) - Y_i(0) - \tau) \quad (4.0.1)$$

$$= \begin{cases} Y_i^{obs} - \alpha & \text{if } W_i = 0, \\ Y_i^{obs} - \alpha - \tau & \text{if } W_i = 1 \end{cases} \quad (4.0.2)$$

$$= Y_i^{obs} - (\alpha + \tau \cdot W_i) \quad (4.0.3)$$

and thus we have

$$Y_i^{obs} = \alpha + \tau \cdot W_i + \varepsilon_i$$

4.1 Unbiasedness of the Least Squares Estimator

$$\mathbb{E}[\hat{\beta}|W_1, \dots, W_N] = \mathbb{E}\left[\frac{\sum_{i=1}^N (W_i - \bar{W})(Y_i - \bar{Y})}{\sum_{i=1}^N (W_i - \bar{W})^2} \mid W_1, \dots, W_N\right] \quad (4.1.1)$$

$$= \mathbb{E}\left[\frac{\sum_{i=1}^N (W_i - \bar{W})(Y_i - (\alpha + \beta\bar{W} + \bar{\varepsilon}))}{\sum_{i=1}^N (W_i - \bar{W})^2} \mid W_1, \dots, W_N\right] \quad (4.1.2)$$

$$= \mathbb{E}\left[\frac{\sum_{i=1}^N (W_i - \bar{W})(\varepsilon_i - \bar{\varepsilon} + \beta(W_i - \bar{W}))}{\sum_{i=1}^N (W_i - \bar{W})^2} \mid W_1, \dots, W_N\right] \quad (4.1.3)$$

$$= \beta + \frac{\frac{1}{N} \sum_{i=1}^N (W_i - \bar{W}) \mathbb{E}[\varepsilon_i - \bar{\varepsilon} | W_1, \dots, W_N]}{\frac{1}{N} \sum_{i=1}^N (W_i - \bar{W})^2} = \beta \quad (4.1.4)$$

Note that the denominator in equation (4.1.4) is only defined if $\sum_{i=1}^N W_i \in (0, N)$, and is equal to $\frac{M}{N}(1 - \frac{M}{N})$. The numerator is $\frac{M}{N}(1 - \frac{M}{N})(\bar{Y}_T - \bar{Y}_C)$. The bottom line is, therefore, $\hat{\beta} = \bar{Y}_T - \bar{Y}_C = \hat{\tau}$.

4.2 Assumptions of each approach

4.2.1 Neyman approach

- We have a finite population, we only see one of the potential outcomes for each unit.
- The source of randomness is the permutation of the treatment indicator.
- The unbiasedness comes from $\mathbb{E}[\bar{Y}_T - \bar{Y}_C | Y_1(1), Y_1(0), \dots, Y_N(1), Y_N(0)] = \frac{1}{N} \sum_{i=1}^N (Y_i(1) - Y_i(0))$

4.2.2 LS approach

- Model: $Y_i = \alpha + \beta W_i + \varepsilon_i$
- Assumption: $\varepsilon_i \sim N(0, \sigma^2)$ and $\varepsilon_i \perp W_i$
- The unbiasedness comes from the independence: $\mathbb{E}[\hat{\beta} | W_1, \dots, W_N] = \beta$

Which assumptions are more plausible and how should we judge that plausibility? Suppose I want to compare the change in employment in NJ fast food restaurants to the change in employment in PA restaurants, as in Card and Krueger(1996; AER). What is a good way to think about the two approaches? We can analyze the randomized experiment from a *design perspective* as in the Neyman approach; or we can also analyze it using the least squares approach, which is based on a *sampling perspective*. In practice, these two lead to same/very-similar conditions, but it takes a quite different perspective on the interpretation of the uncertainty. Recall that both estimators are the exact same in terms of how they are constructed. In the next section,

4.3 The Variance of the Least Squares Estimator

The least squares estimator is given by:

$$\hat{\beta} = \frac{\sum_{i=1}^N (W_i - \bar{W})(Y_i - \bar{Y})}{\sum_{i=1}^N (W_i - \bar{W})^2} = \beta + \frac{\frac{1}{N} \sum_{i=1}^N (W_i - \bar{W}) \varepsilon_i}{\frac{1}{N} \sum_{i=1}^N (W_i - \bar{W})^2} \quad (4.3.1)$$

This is a linear combination of the error terms, assumed to be iid normal random variables. The distribution of the least squares estimator is given by:

$$\hat{\beta} \sim N\left(\beta, \frac{\sigma^2}{N \frac{N_1}{N} \left(1 - \frac{N_1}{N}\right)}\right) \quad (4.3.2)$$

Therefore, the variance of the least squares estimator is given by:

$$\mathbb{V}(\hat{\beta}) = \frac{\sigma^2}{N \frac{N_1}{N} \left(1 - \frac{N_1}{N}\right)} \quad (4.3.3)$$

which can be also written as:

$$\mathbb{V}(\hat{\beta}) = \frac{N\sigma^2}{N_1(N - N_1)} = \frac{\sigma^2}{N_1} + \frac{\sigma^2}{N - N_1} \quad (4.3.4)$$

4.3.1 The Estimated Variance of the Least Squares Estimator

Given the residuals of $\hat{\varepsilon}_i = Y_i - \hat{\alpha} - \hat{\beta}W_i$, we can estimate the variance of the error terms as:

$$\hat{\sigma}^2 = \frac{1}{N - 2} \sum_{i=1}^N \hat{\varepsilon}_i^2 \quad (4.3.5)$$

Therefore, the estimated variance of the least squares estimator is given by:

$$\hat{\mathbb{V}}(\hat{\beta}) = \frac{\hat{\sigma}^2}{N \frac{N_1}{N} \left(1 - \frac{N_1}{N}\right)} = \frac{\frac{1}{N-2} \sum_{i=1}^N \hat{\varepsilon}_i^2}{N \frac{N_1}{N} \left(1 - \frac{N_1}{N}\right)} \quad (4.3.6)$$

$$= \frac{\frac{1}{N-2} \sum_{i=1}^N \hat{\varepsilon}_i^2}{N_1} + \frac{\frac{1}{N-2} \sum_{i=1}^N \hat{\varepsilon}_i^2}{N - N_1} \quad (4.3.7)$$

4.4 Comparison of the variance results in Neyman and OLS

Now, compare the OLS variance results to the Neyman variance results:

	Neyman	OLS
Variance	$\frac{1}{N_1} S_1^2 + \frac{1}{N_0} S_0^2 - \frac{1}{N} S_{01}^2$	$\frac{\sigma^2}{N \frac{N_1}{N} \left(1 - \frac{N_1}{N}\right)}$
Estimated Variance	$\frac{1}{N_1} s_1^2 + \frac{1}{N_0} s_0^2$	$\frac{\hat{\sigma}^2}{N_1} + \frac{\hat{\sigma}^2}{N - N_1}$

Table 2: Comparison of the Neyman and OLS variance estimators

4.4.1 Why are the true variances different?

This is because we have a different set of assumptions:

- **Neyman:** $W_i \perp \{Y_i(0), Y_i(1)\}$
- **OLS:** $W_i \perp \varepsilon_i$ and $\varepsilon_i \sim N(0, \sigma^2)$, followed by

$$Y_i = \alpha + \beta W_i + \varepsilon_i$$

4.4.2 Why are the estimated variances different?

This is because we have a different set of assumptions: (TBU)

5 Variance Estimation using Bootstrap and Resampling Methods

5.1 Variance Estimation under Randomization

Upto now, we have been focusing on the estimator $\hat{\tau} = \bar{Y}_T - \bar{Y}_C$ that is unbiased under randomization. Depending on the source of randomness, we had two different variance definitions:

$$\mathbb{V}_{\text{Neyman}}(\hat{\tau}) = \frac{1}{N_1} S_1^2 + \frac{1}{N_0} S_0^2 - \frac{1}{N} S_{01}^2 \quad (5.1.1)$$

$$\mathbb{V}_{\text{OLS}}(\hat{\tau}) = \frac{\sigma^2}{\sum_{i=1}^N (W_i - \bar{W})^2} = \frac{\sigma^2}{N \frac{N_1}{N} (1 - \frac{N_1}{N})} \quad (5.1.2)$$

In the Neyman variance, S_1^2 and S_0^2 are the population⁹ variances of $Y_i(1)$ and $Y_i(0)$, respectively; the remaining term S_{01}^2 is **the population variance of the unit-level treatment effect**. In the OLS variance, σ^2 is the population variance of the normal error term (ε_i), and $\bar{W}$ is the mean of the treatment indicator among our experimental units.

The corresponding estimators for (5.1.1) and (5.1.2) are given by:

$$\hat{\mathbb{V}}_{\text{Neyman}}(\hat{\tau}) = \frac{1}{N_1} \hat{S}_1^2 + \frac{1}{N_0} \hat{S}_0^2 \quad (5.1.3)$$

$$\hat{\mathbb{V}}_{\text{OLS}}(\hat{\tau}) = \frac{\hat{\sigma}^2}{\sum_{i=1}^N (W_i - \bar{W})^2} = \frac{\hat{\sigma}^2}{N \frac{N_1}{N} (1 - \frac{N_1}{N})} \quad (5.1.4)$$

where $\hat{S}_1^2, \hat{S}_0^2$ are the sample variances of the observed outcomes in the treated and the control groups, respectively; $\hat{\sigma}^2 = \frac{1}{N-2} \sum_{i=1}^N \hat{\varepsilon}_i^2$ is the unbiased estimator of σ^2 .

Additionally, allowing *heteroscedasticity* in the normal error term, we can derive the following variance estimand and the corresponding estimator:

$$\mathbb{V}_{\text{robust}}(\hat{\tau}) = \frac{1}{(N\bar{W}(1-\bar{W}))^2} \sum_{i=1}^N \left(\sigma_i^2 (W_i - \bar{W})^2 \right) \quad (5.1.5)$$

$$\hat{\mathbb{V}}_{\text{robust}}(\hat{\tau}) = \frac{1}{(N\bar{W}(1-\bar{W}))^2} \sum_{i=1}^N \left(\hat{\varepsilon}_i^2 (W_i - \bar{W})^2 \right) \quad (5.1.6)$$

In this section, we discuss additional class of variance estimators based on **resampling methods**. By the end of this section, you will be able to:

- Understand the intuition behind resampling methods.
- Understand the sense of richness of these ideas.
- Understand how to implement the methods in practice.

⁹Note that the Neyman world views the experiment participants as the population, and the source of randomness is the permutation of the treatment indicator.

5.2 One-Sample Example

Suppose we have a random sample $Y_1, \dots, Y_N$ from a distribution with mean μ and variance σ^2 , and we want to estimate the population mean μ . This is a simpler problem than estimating the average treatment effect τ in the randomized experiment, but it still conveys the essence of the resampling methods. The natural estimator of μ is the sample mean $\bar{Y} = \frac{1}{N} \sum_{i=1}^N Y_i$. The variance of this estimator is given by $\mathbb{V}(\bar{Y}) = \frac{\sigma^2}{N}$, which we typically estimate with:¹⁰

$$\hat{\mathbb{V}}(\bar{Y}) = \frac{\hat{\sigma}^2}{N} = \frac{1}{N-1} \sum_{i=1}^N (Y_i - \bar{Y})^2 \quad (5.2.1)$$

Here's the key point: **The variance of any function of N random variables can be characterized in terms of the joint distribution of those random variables**, where the distribution can be written in terms of, CDFs($F_Y(y) = Pr(Y \leq y)$) or moment generating functions($M_Y(t) = \mathbb{E}(e^{tY})$), for example. In our case with a *random sample*, 5.2.1 can be written as:

$$\frac{\hat{\sigma}}{\sqrt{N}} = \frac{1}{N} \quad (5.2.2)$$

where $\bar{Y}$ is the sample mean of the random sample $Y_1, \dots, Y_N$.

5.3 Bootstrap Method(TBU)

¹⁰The OLS variance estimator is essentially a generalization of this estimator to the regression case, but in the mean case we do not have to worry about heteroscedasticity.

6 More Regression Stuff

In this section, we consider mainly the following dataset:

- Lalonde Data from Dehejia and Wahba (1999)
- Evaluation of the labor market program in 1968(i.e., treatment date is 1968)
- It has two datasets:
 - ◊ **Experimental**: randomized treatment assignment to the labor market program(185 T & 260 C)
 - ◊ **Observational(CPS)**: control group from the CPS survey (185 T & 15,992 C)
- Outcome Variable: Earnings in 1978(two yrs after the treatment)
- Other Variables:
 - ◊ Program Participation
 - ◊ Age
 - ◊ Earnings in 1974/1975
 - ◊ Children
 - ◊ Education
 - ◊ Ethnicity
 - ◊ etc.

The main question is *whether the labor market program is effective in increasing the earnings of the participants*.

Consider the following four sets of regressions:

Table 3: Regression Models

Outcome	Data	Intercept	Program	R ²
Earnings(1978)	Experimental	4.55(0.34) ^{***}	1.79(0.67) ^{***}	0.02
	Observational(CPS)	14.85(0.08) ^{***}	0.68(0.03) ^{***}	0.01
Unemployment(1975)	Experimental	0.68(0.03) ^{***}	-0.08(0.05)	0.01
	Observational	0.109(0.003) ^{***}	0.49(0.04) ^{***}	0.03

References

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- Dehejia, Rajeev H. and Sadek Wahba (1999). “Causal Effects in Nonexperimental Studies: Reevaluating the Evaluation of Training Programs”. In: *Journal of the American Statistical Association* 94.448, pp. 1053–1062.